Homework 9

Problem 1

In roulette, a bet on a single number has a 1/38 probability of success and pays 35-to-1. That is, if you bet 1 dollar, your net winnings are -1 with probability 37/38 and +35 with probability 1/38. Consider betting on a single number on each of n spins of a roulette wheel. Let $\bar{X}_n$ be your average net winnings per bet.

For each of the values $n = 10$, $n = 100$, $n = 1000$:

1. Compute $\mathrm{E}(\bar{X}_n)$
2. Compute $\mathrm{SD}(\bar{X}_n)$
3. Run a simulation to determine if the distribution of $\bar{X}_n$ is approximately Normal. (It's fine if you use an applet, but you can also try coding)
4. Use simulation to approximate $\mathrm{P}(\bar{X}_n > 0)$, the probability that you come out ahead after n bets.
5. If $n = 1000$ use the Central Limit Theorem to approximate $\mathrm{P}(\bar{X}_{1000} > 0)$, the probability that you come out ahead after 1000 bets.
6. The casino wants to determine how many bets on a single number are needed before they have (at least) a 99% probability of making a profit. (Remember, the casino profits if you lose; that is, if $\bar{X}_n < 0$.) Use the Central Limit Theorem to determine the minimum number of bets (keeping in mind that n must be an integer). You can assume that whatever n is, it's large for the CLT to kick in.

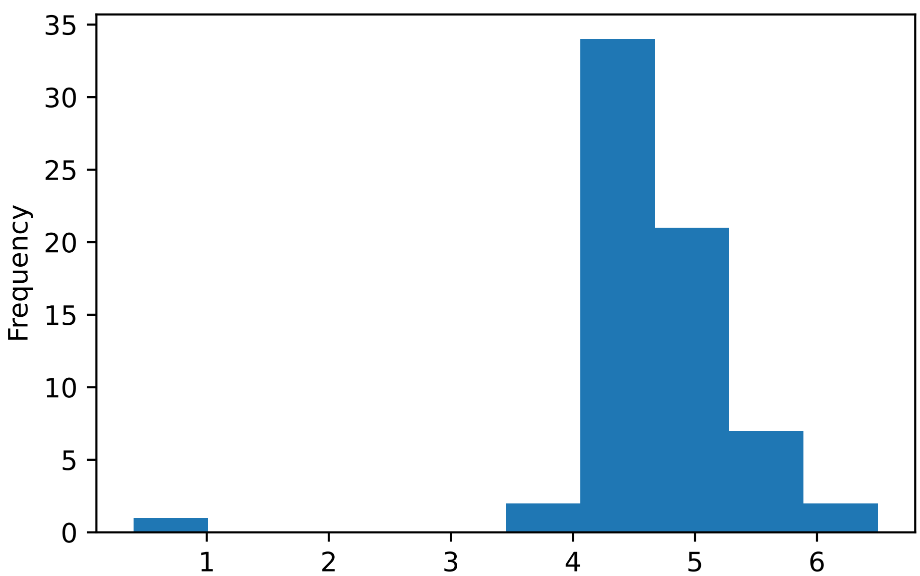
Problem 2

The standard measurement of the alcohol content of drinks is alcohol by volume (ABV), which is given as the volume of ethanol as a percent of the total volume of the drink. In a sample of 67 brands of beer, the mean ABV is 4.61 percent and the standard deviation of ABV is 0.754 percent.

We should never analuze data without first looking at some plots, but we're going to that now.

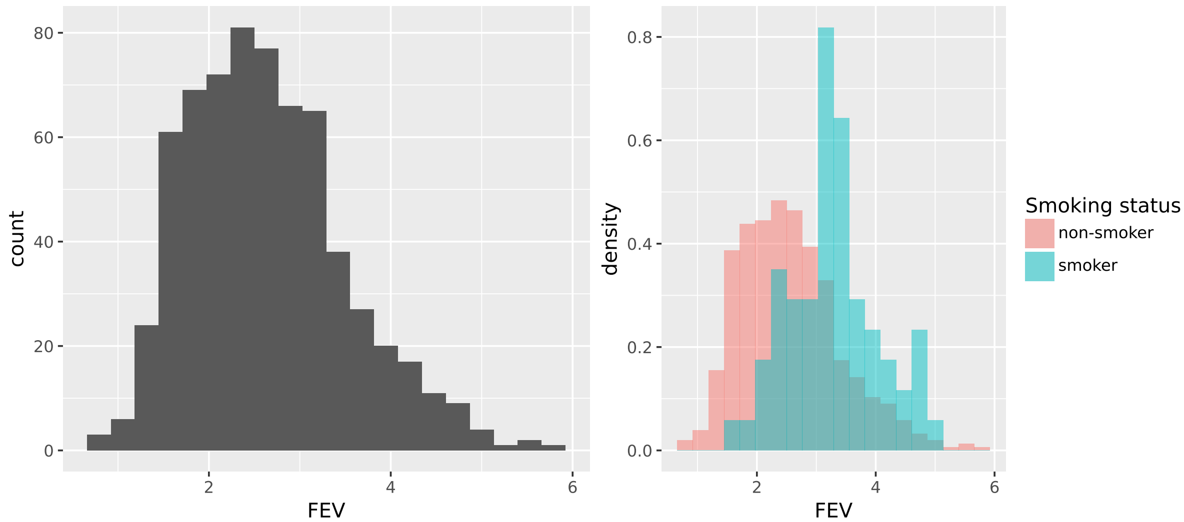
1. Compute a 95% confidence interval for the appropriate population mean.

2. Write a clearly worded sentence reporting your confidence interval in context.
3. Is 4.5% a plausible value of the parameter? Explain briefly.
4. When we plot the data (which we should have done first!) we notice an outlier (see plot below). One of the brands of beer in the sample is O'Doul's, a non-alcoholic beer. The ABV for O'Doul's is 0.4% (it has a bit of alcohol.) Suppose O'Doul's is removed from the data set. Compute the sample mean ABV of the remaining 66 brands.
5. The sample SD of ABV of the remaining 66 brands is 0.55 percent. Explain intuitively why this value is smaller than the sample SD of all 67 brands.
6. Compute the 95% confidence interval based on the sample with O'Doul's removed. Compare to the original interval, both in terms of center of the CI and its width. Explain briefly.
7. Based on the interval based on the sample with O'Doul's removed, is 4.5% a plausible value of the parameter? Explain briefly.
8. Which of the analyses is more appropriate: with or without O'Doul's? Explain your reasoning.



Problem 3

Do non-smokers have better lungs than smokers? One measure of lung function is forced expiratory volume (FEV), the amount of air (in liters) an individual can exhale in the first second of forceful breath. Larger values of FEV are indicative of better functioning lungs. In a study, FEV was measured for a group of 654 subjects, along with whether they smoked and other variables like age (years). Note: I'm purposely not yet giving you much information about how the sample was collected.



(a) (b)

1. Suppose you want to fill in the blanks in the following sentence: 95% of FEV values in the population are between [blank] and [blank]. Assuming this sample is representative, provide reasonable values that fill in the blanks, and explain your reasoning. Hint: it is better to give a correct rough estimate than a precise but incorrect answer, so think before trying to compute anything. (Since you don't have much information about the sample, you don't have to clarify yet what "population" is appropriate.)
2. Using appropriate summary statistics, compute by hand a 95% confidence for the appropriate difference in population means.
3. Write a clearly worded sentence interpreting the confidence interval in context.
4. Assuming this is a representative sample, are we reasonably confident that one group (smokers or non-smokers) tends to have better FEV? Why? Which group?
5. You should have observed something surprising in the previous parts. You decide to look closer at the data (you should have done this in the first place) and discover that the 654 subjects in this study were all children! Their ages varied from 3 to 19 years old. How does this change your conclusions? Can you suggest an explanation for the surprising result?

Problem 4

The following table displays the time (measured continuously in minutes) until the first goal was scored in each of 20 professional hockey games. The sample mean is 12.1 minutes, the sample median is 8.9 minutes, and the sample SD is 13.1 minutes.

4.4	14.9	2.8	1.4	1.1	8.1	11.7	3.9	2.4	15.7
8.8	0.6	5.1	10.4	9.1	13.7	27.8	9.0	46.0	44.7

1. You wish to estimate the population *median* time until the first goal is scored. Describe in detail in words how you could use the sample data and simulation to find an appropriate 95% bootstrap percentile confidence interval.
2. The following summarizes the bootstrap distribution of the sample median based on the sample data and an appropriate simulation.

Min	2.5th percentile	Median	97.5th percentile	Max	Mean	SD
1.25	4.15	8.90	12.70	19.75	8.52	2.10

   Using the above information, compute the endpoints of each of the following bootstrap 95% confidence intervals for the population median.

   a. Normal interval
   b. Percentile interval

3. Write a clearly worded sentence reporting the bootstrap percentile confidence interval from the previous part in context.

Problem 5

Two different machines (A and B) that fill packages of a certain candy are calibrated to a weight of 50 grams. Naturally, the weights of individual packages vary somewhat in the production process, but too much variation is undesirable. A small sample of packages is taken from each machine; the weights (grams) are in the following table.

Machine						Mean	SD
A	47.1	48.7	50.1	50.2	50.5	49.3	1.4
B	48.6	50.5	50.6	51.4	52.0	50.6	1.3

1. Describe in detail in words how you could use the sample data and simulation to find a 95% bootstrap percentile confidence interval for the *ratio of the variances* of packages weights for the two machines.
2. The confidence interval from the previous part is (0.01, 14.4). Write a clearly worded sentence reporting the confidence interval from the previous part in context.
3. The confidence interval was based on 10000 bootstrap samples. Explain why the interval is so imprecise.